29TH INTERNATIONAL MARS SOCIETY CONVENTION

THE INTELLIGENCE LAYER FOR MARS

Harnessed agentic AI for science, engineering, space transportation, communications and settlement



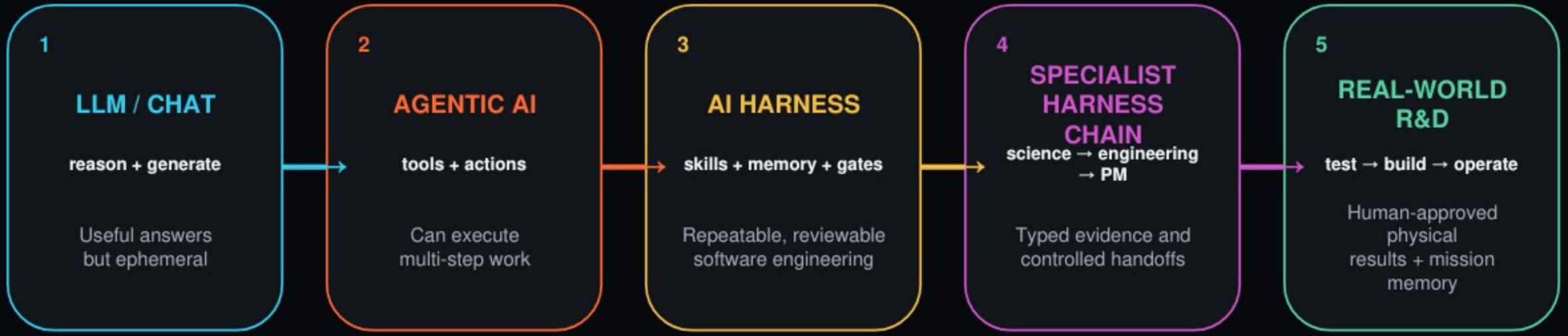


OCTOBER 22–24, 2026 • USC, LOS ANGELES

BOB J MILLS

FROM CHAT TO REAL-WORLD R&D

The breakthrough is not a smarter chat window. It is a governed chain of specialist harnesses that turns AI reasoning into durable work.



Claude Code / Codex show the middle transition: AI becomes an engineering worker. The Mars harness extends that pattern into science, design, verification, project execution and operations.

HUMANS SET INTENT • REVIEW EVIDENCE • APPROVE RELEASE • OWN RISK

COMPRESS THE NON-PHYSICAL PART OF R&D

Automate coordination, research synthesis, documentation and rework propagation—while keeping experiments, expert judgment and approval human-governed.

TRADITIONAL R&D LATENCY

Search + synthesize	weeks → hours
Draft + document	weeks → hours
Cross-discipline handoffs	days/weeks → minutes
Trace changes + rework	days → automated propagation
Build review packages	days → hours

PROGRAM THESIS / TARGET IMPACT

YEARS → DAYS

for selected concept-development and documentation loops

$MILLIONS

potentially avoided in repeated engineering effort, coordination delay and late discovery of inconsistencies

NOT "REMOVE THE HUMANS." REMOVE LOW-VALUE LABOR SO HUMANS CAN SPEND MORE TIME ON SCIENCE, ENGINEERING JUDGMENT, TESTING AND DECISIONS.

Mars is a learning-rate problem

The scarcest resource is time - from a goal to execution.

The critical path is validated science → controlled engineering → authorized execution, with evidence and blockers moving through typed gates.



Research And Development

Project Management

A Ai agentic harness attacks coordination latency while preserving the evidence needed to close decisions.

An LLM proposes. The harness controls what can become mission state.

The model is probabilistic. The harness supplies identity, state, tools, typed contracts, independent validation, memory, and human authority.

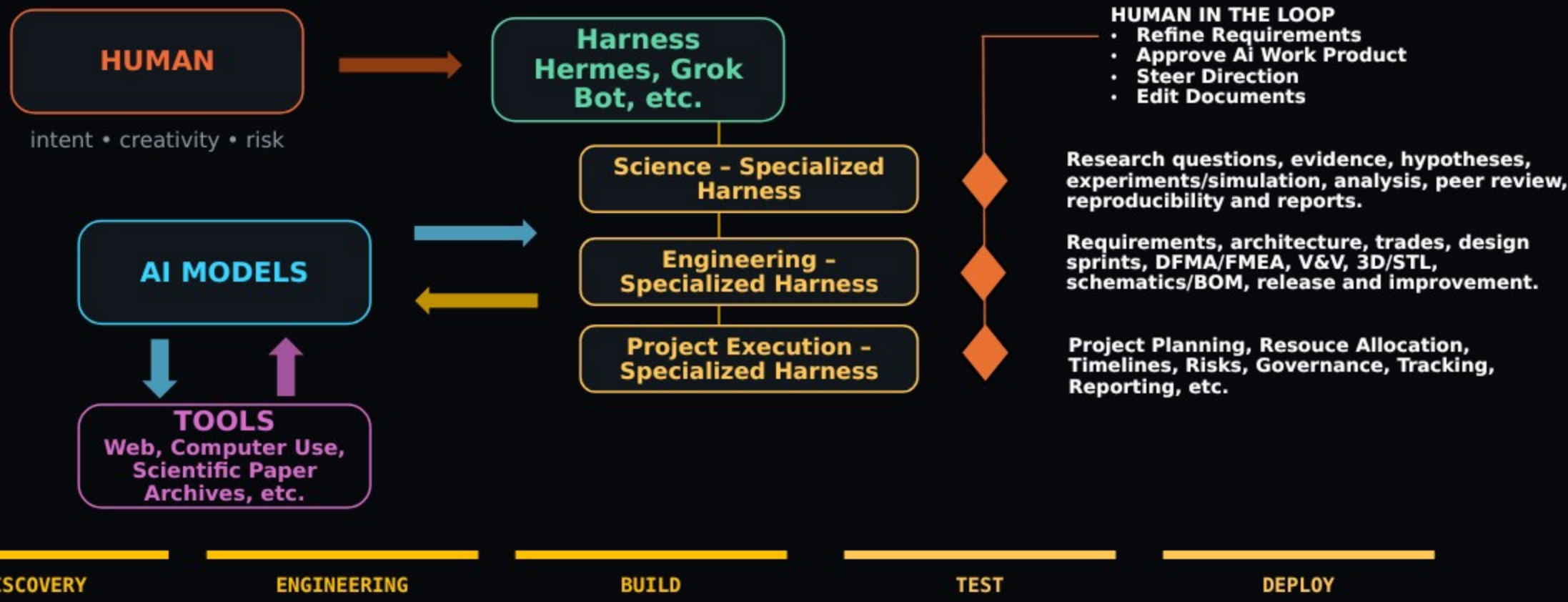




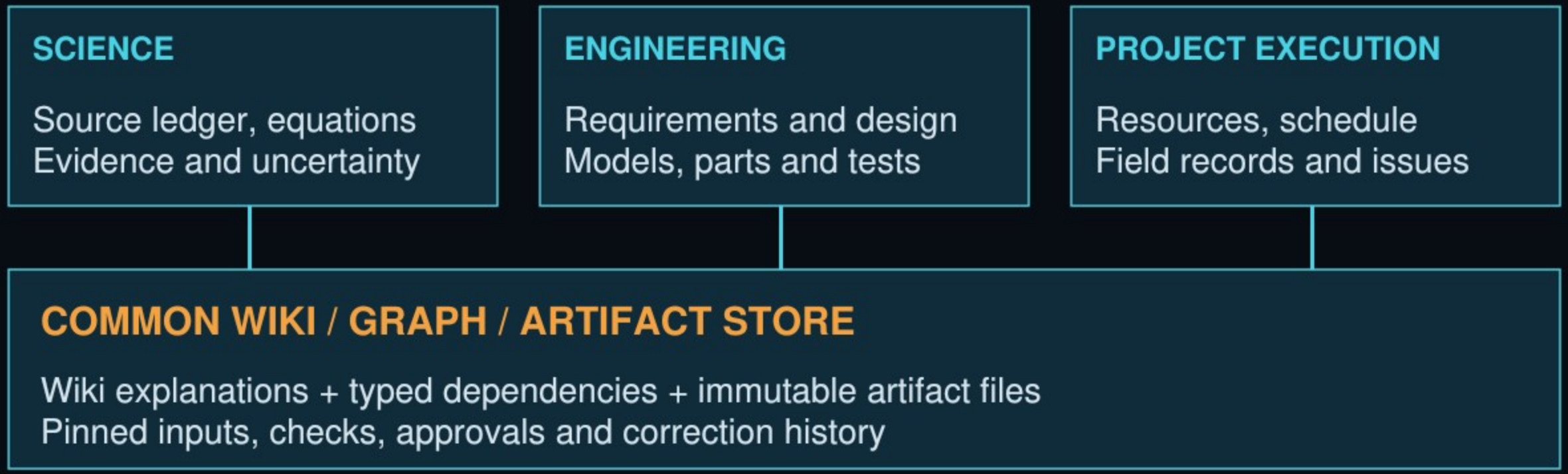
Stable Skills, Contracts and Consistent Approval authority let every stronger future Ai model inherit a mission-ready way to continue the work.

Skills & Agent identity are durable

Define the capability, evidence, permissions and work product first. Then assign the best executor available.



Specialized harnesses share a common artifact graph



Approved outputs flow forward. Targeted correction requests can return upstream.

First principles reveals better design options

1 ASSUMPTIONS

Law, measured fact, estimate, habit or opinion

2 FUNDAMENTALS

Dated sources, units, bounds and uncertainty

3 COST GAP

Current baseline versus a feasible lower bound

4 REBUILD

Bold redesign and conservative alternative

5 STRESS TEST

First failure, skeptical challenge and test

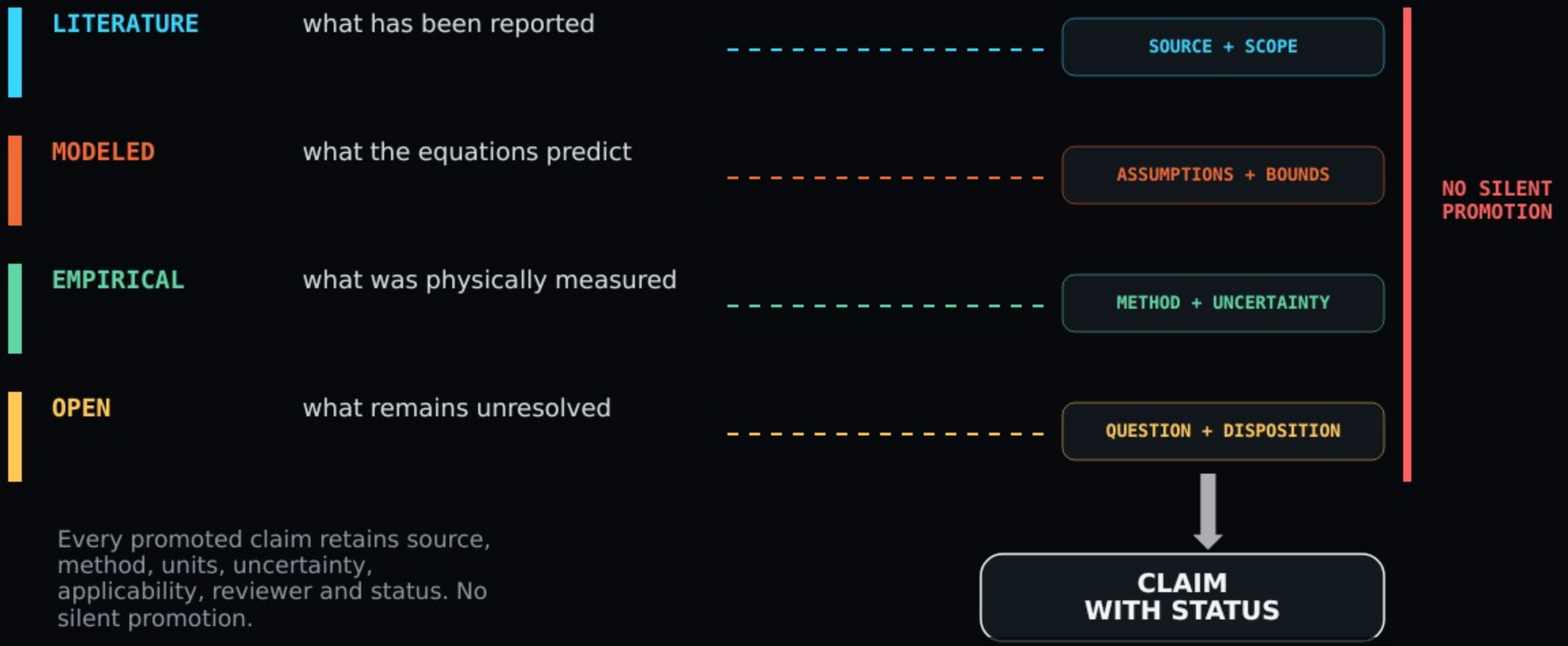
6 DECISION

Best supported option and first reversible step

Deliver assumptions, source ledger, cost-gap table and a 30-day plan.

Every claim carries epistemic status

The harness prevents a plausible sentence—or a clean simulation—from silently becoming a physical fact.



Every promoted claim retains source, method, units, uncertainty, applicability, reviewer and status. No silent promotion.

Intelligent Engineer turns approved evidence into a controlled technical baseline

Drafts accelerate work. Phase gates authorize work.
Automated Engineering with Human in the loop



REQUIRE	TRADE	DESIGN + SPRINTS + SPRINTS	DFMA / FMEA	VERIFY	LAUNCH / OPERATE
mission + constraints	architecture + margins	PDR / CDR sprints	DFMA • FMEA • faults	methods + acceptance	versioned baseline

LOCKED BASELINE

Only reviewed phase outputs become authorized inputs downstream.



Every action returns a checkable receipt

CONTRACT

Goal, inputs and acceptance

ACTION

One bounded tool step

RECEIPT

Result, paths and error

VERIFY

Evidence against criterion

PASS

Evidence satisfies the check

FAIL

Evidence contradicts it

BLOCKED

A dependency prevents it

UNVERIFIED

Evidence is insufficient

Repair within limits. Save a checkpoint. Deliver artifacts with evidence.

Downstream blockers trigger controlled regeneration

DETECT	REQUEST	REGENERATE	REVALIDATE
Failed check or execution blocker	Pinned revisions and bounded question	Upstream owner creates a revision	Rebuild dependents and check release

B1 stays in history. Correction branch C1 holds revised work.

B2 becomes current only after approved upstream evidence, downstream checks and required authority agree.

Stale inputs block promotion. Unrelated work needs a recorded independence check.

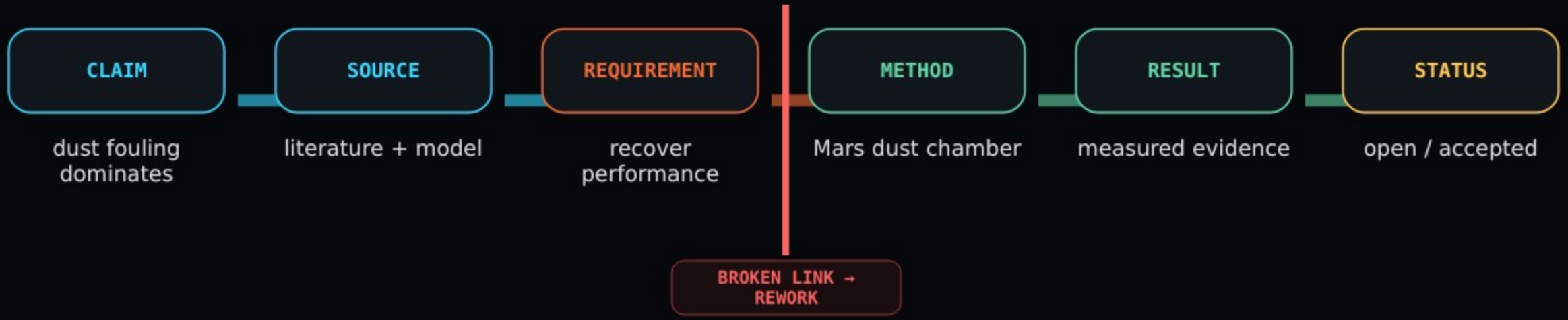
Recursive specialists shorten waiting; bounded gates preserve truth

Eleven skill-bound agents form the control spine. Domain directors can spawn narrower specialists under typed contracts that cannot widen permissions, budget or authority.



Ground truth starts at the first claim

The Ground Truth Gatekeeper checks provenance and configuration, recomputes critical values, runs tests, and controls promotion independently.



Every uncertainty gets a disposition

ANALYZE | TEST | CONSTRAIN | ACCEPT RISK

No agent is the sole approver of its own mission-critical work. Missing physical evidence remains missing.

Artifact revisions preserve evidence and dependencies

IDENTITY	Stable artifact ID, immutable revision, file location and hash
LINEAGE	Producer run, exact input revisions and baseline/configuration
KNOWLEDGE	Source links, assumptions, evidence class and typed relationships
CONTROL	Draft, reviewed, approved, stale or blocked; accountable owner
PROMOTION	Coherent manifest, independent gates and revision-conflict check

Retain prior bytes and approvals. Supersede explicitly. Redact before persistence.

MOXIE flight evidence informs a modeled scale-up study

SCALED MOXIE

304 kWe* Modeled continuous power

592 t* Modeled O_2 over 730 days

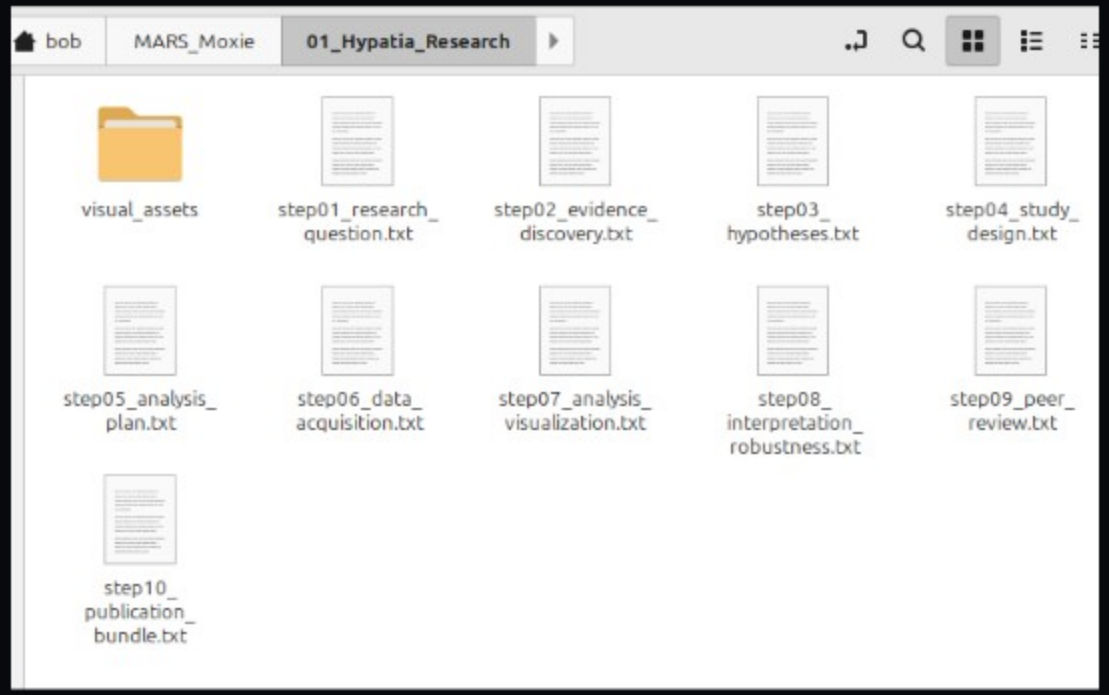
≥98% MOXIE demonstrated purity

Modeled O_2 allocation, tonnes



Science asks: Is settlement-scale MOXIE physically plausible?

Step 10: Publication Bundle - Project ARES-SOE
Title: Industrial-Scale Solid Oxide CO_2 Electrolysis (SOXE) for Human Mars Missions: A 33.8 kg/hr Modular Architecture for Habitat Life Support and Starship Propellant Refueling
Author List: Hypatia AI (Lead Systems Architect), Mars In
1. Abstract As human Mars exploration transitions from exploration to habitation, In-Situ Resource Utilization (ISRU) must scale by three orders of magnitude. This paper presents the design and validation framework for Project ARES-SOE (Advanced Regenerative Electrolysis System for Sustained

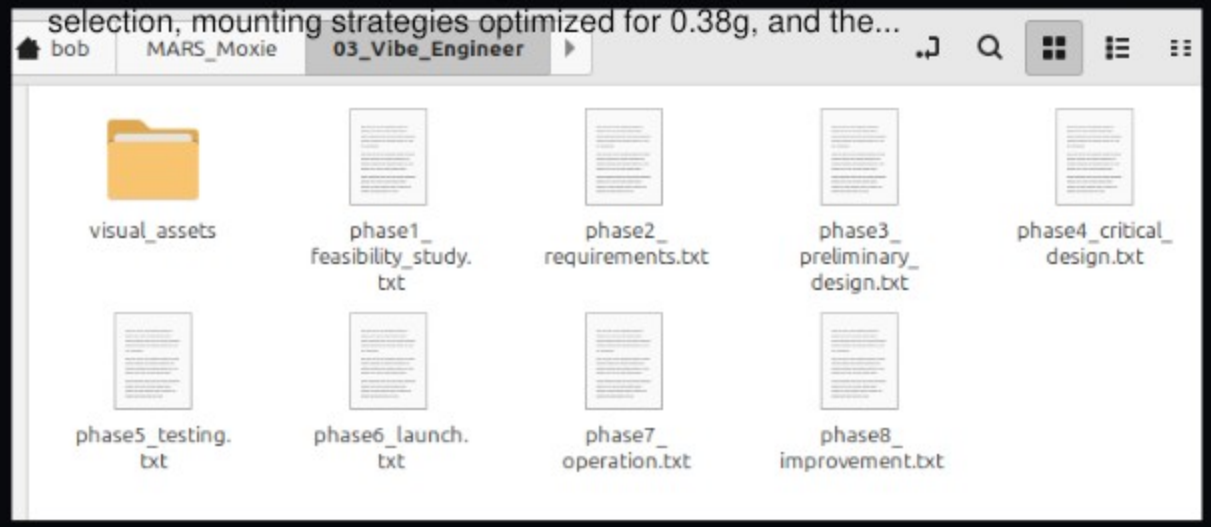


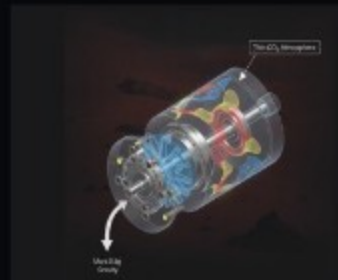
Oxygen Production), a scaled-up solid-oxide CO_2 electrolysis system. Building on the heritage of the NASA MOXIE instrument, the system is designed to produce 592.4 t of O_2 over 2 Earth years (17,520 hours)—sufficien t to sustain a 12-person habitat and provide 50% of



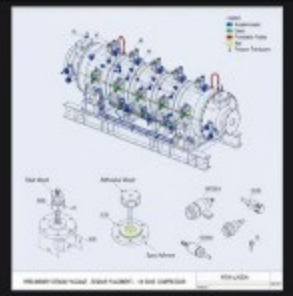
Then the engineering... What would this look like?

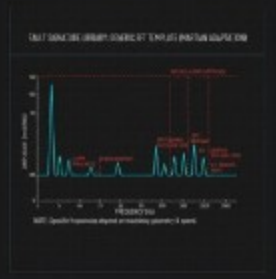
DOCUMENT ID: SOXE-VIB-CDR-004
PROJECT: Mars SOXE (Solid-Oxide Carbon-Dioxide Electrolysis) Scale-Up Plant
PHASE: Critical Design Review (CDR)
AUTHOR: Hypatia | Senior Vibration Lead (VibraEngineer)
DATE: 2024.05.22
SUBJECT: Vibration Monitoring & Structural Integrity Systems for modeled 33.8 kg/h O_2 generation
1. EXECUTIVE OVERVIEW
The SOXE plant represents an 11x scale-up of the IAC-2022 baseline, necessitating a shift from experimental monitoring to industrial-grade autonomous reliability. With CO2 feed compressors drawing the bulk of the modeled 304 kWe power budget, vibration is the primary failure mode for the 17,520-hour continuous campaign. This document outlines the final sensor selection, mounting strategies optimized for 0.38g, and the...




01_feasibility.png
1024x1024 1240KiB


02_requirements.png
1024x1024 1224KiB


03_preliminary_design.png
1024x1024 1524KiB

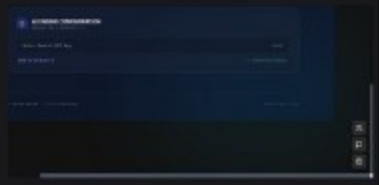

04_critical_design.png
1024x1024 1454KiB


05_testing.png
1024x1024 1701KiB


06_launch.png
1024x1024 1673KiB

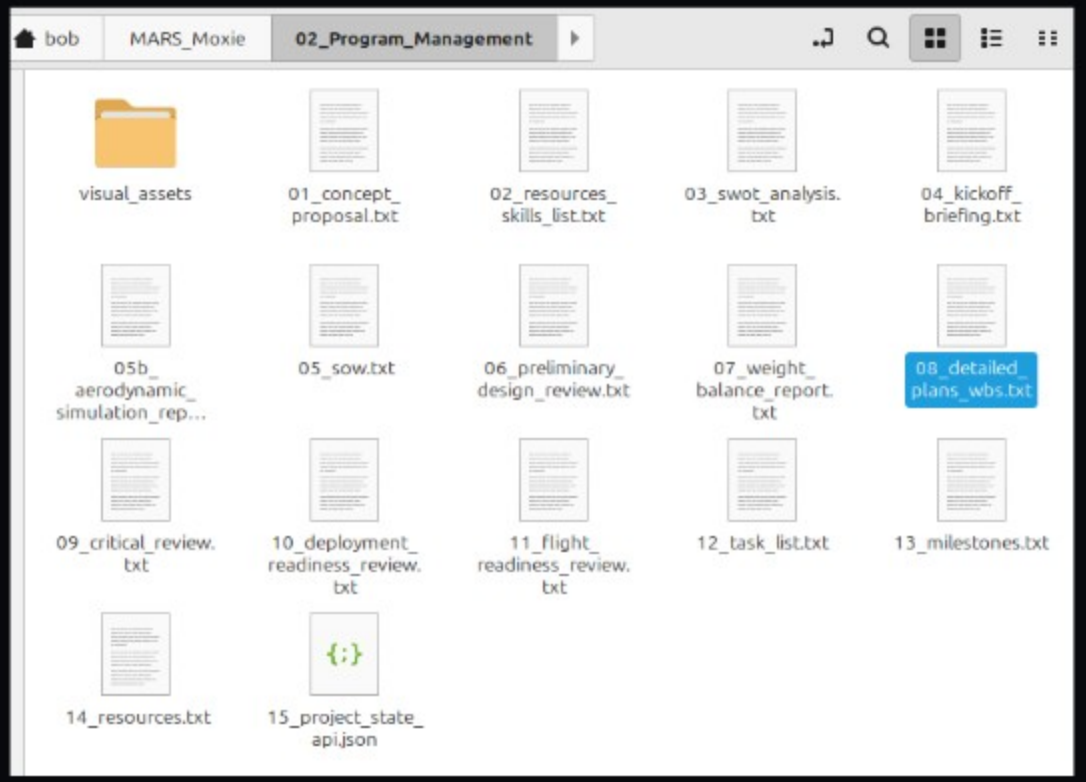

07_operation.png
1024x1024 1551KiB


08_improvement.png
1024x1024 1067KiB


vibe_engineer_dashboard.png
1280x577 54KiB

Then a project plan… Resources, Costs, Time, Tracking?

WBS
1.0 Atmospheric acquisition + dust rejection
2.0 Solid-oxide CO_2 electrolysis stacks
3.0 Oxygen cleanup, liquefaction + storage
4.0 Modeled 304 kWe power infrastructure



5.0 Integration + mission operations
6.0 Verification + qualification
7.0 Deployment, commissioning + tracking



The harness can accelerate an integrated first-pass concept—not substitute for qualification.



A design or plan-blocking uncertainty becomes an executable Research Request Package—not a vague demand for more study.

- AI interacts between all steps feeding each step context to the next
- AI-driven change management feeds back and re-executes previous steps upgrading documentation
- Every step has its own agentic harness designed to deliver intermediate documents and provide the downstream steps with input
- Every step can run autonomously or can be manually reviewed and edited step by step

HYPOTHESES	REQUIREMENTS	PLAN STEPS
RESEARCH	PRELIM DESIGN	RESOURCES
EXPERIMENT	DESIGN SPRINTS	TIMELINES
ANALYSIS	FINAL DESIGN	PROJECT PLAN
VALIDATION	VALIDATION	TRACK PROGRESS

Five controls keep accelerated work mission-credible



Typed packets preserve context. Evidence classes prevent false promotion. Independent gates control release. Mission memory preserves state. Blockers return to research.

Scale-up is plausible	System integration dominates	WBS connects all systems
Evidence classes must stay explicit	304 kWe remains modeled	Cost + schedule need basis
33.8 kg/h remains modeled	Requirements need traceability	Risks remain visible
Degradation needs experiments	Reliability needs qualification	Changes propagate through state
Open questions drive research	Design must return blockers	Approval gates control release

Workflows become portable mission skills

Codex, Claude Cowork, Hermes and Grok and their future versions can provide the agentic workflows needed captured as Skills and Agent Idenities. Below is a small sample of where this could be beneficial.



Reviewed lessons improve the next working loop

tasks/todo.md

Goal and bounded plan
Observable acceptance checks
Progress and final review

tasks/lessons.md

Mistake and root cause
Scoped prevention rule
Early detection signal

Resume from checked work. Re-plan after evidence changes. Keep workers bounded.

97 requirements mapped. Local checks support the methods. Live host enforcement still needs evidence.

MAKE ENGINEERING A LEARNING SYSTEM

Mars will reward teams that learn faster without losing discipline.



Build the intelligence layer now—so every stronger model inherits mission-ready skills, typed contracts, ground-truth gates and durable state.

The Beginning – Not the End



- The speed of Improvement of Ai is exponential and tomorrow it will be able to do much more than today. But the need for specialized skills and agent identities will still exist.
- Skills can be extended as new mission needs appear, but every new capability should inherit the same evidence, ethics, permissions, review and stop conditions.
- Mars Harness Lab V5.1 publishes the architecture, source review, eleven skills, eleven paired agents, recursive specialist model, engineering discipline catalog, memory starter, case study and downloadable packages under Apache-2.0.
- More information and code are available on the https://www.mifeco.com website.
- I can be contacted at mifecoinc@gmail.com

Yes, this works on Mars, the Moon, the Space Economy and on Earth but I am most interested in accelerating the way to Mars.

Currently we are on the brink of AGI (Artificial General Intelligence) This is where Ai is more capable than all humans in every subject including knowledge (done) and reasoning (close). We need to work hard to stay in the loop and not be overwhelmed. This is a tsunami that will not slow down or stop. I hope I can help in a small way using Ai to keep humans in the loop while solving the BIG issues to make the world (and out of this world) a better place